

EE4410 CYBER SECURITY OF POWER GRIDS

Assignment 3: Cyber-Physical Data Correlation

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[Link to Colab Notebook](#)

1 Familiarization and data visualization

The CPS dataset was loaded and analyzed to understand the main types of signals it contains and how they behave when things are working correctly versus when problems occur. The dataset consists of two types of signals:

- **Cyber signals:** represented by the `numpckt` variable, which records the number of GOOSE packets transmitted every 100 ms.
- **Physical signals:** represented by voltage values from buses 4 to 9, sampled during events. These continuous signals reflect the electrical behavior of the power system under operational or disturbed conditions.

When analyzing these signals, it becomes clear that there is both structure and correlation present, particularly under normal conditions. The GOOSE traffic during normal operation shows a regular, repeating burst pattern with values between zero and three packets per interval, indicating scheduled message exchanges between devices (see Figure 1).

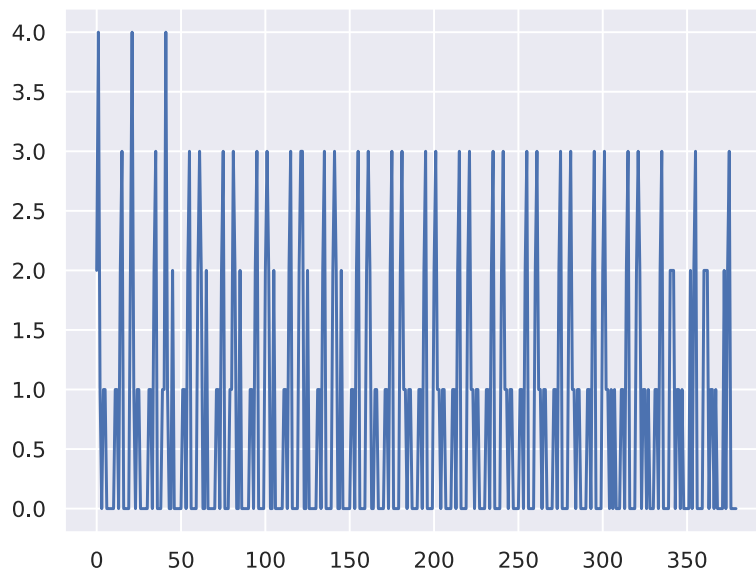
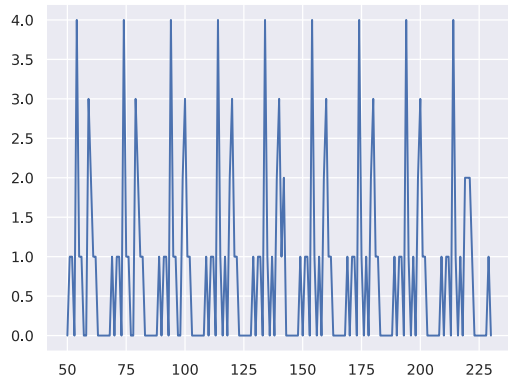
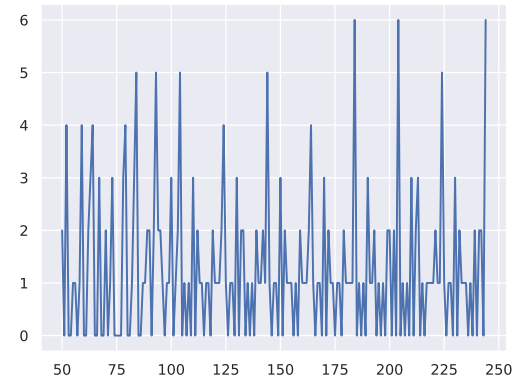


Figure 1: Normal GOOSE traffic

Event A traffic becomes slightly denser but still maintains a semi-periodic form, while Event B traffic is highly irregular and bursty, with peak values exceeding six packets per interval, as illustrated in Figure 2.



Event A GOOSE traffic

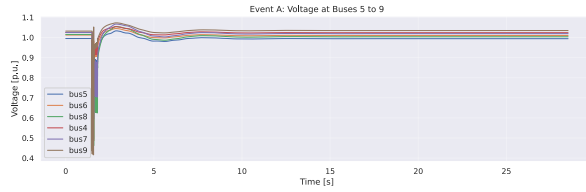


Event B GOOSE traffic

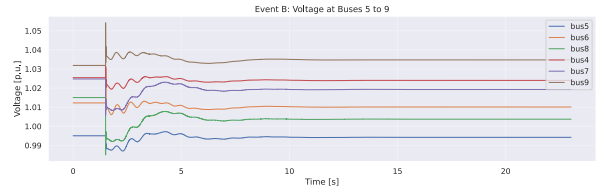
Figure 2: GOOSE traffics plots of Events A and B

This lack of structure in Event B traffic suggests a more chaotic or artificial communication pattern, which may be associated with spoofed or compromised messages.

The voltage plots are shown in Figure 3. During Event A, all monitored buses experience a sudden and dramatic voltage dip to around 0.4 p.u. at approximately $t = 2$ seconds, followed by a swift recovery and steady stabilization by $t = 6$ seconds. This is characteristic of a short-circuit fault that was quickly cleared by protection systems. Event B, by contrast, shows much milder voltage variations. While there is a clear disturbance, it does not involve a sharp drop but rather a series of low-amplitude oscillations. This pattern is consistent with a transient switching operation or an artificial event introduced in the system, possibly through the cyber layer.



Voltage at buses 5–9 during Event A



Voltage at buses 5–9 during Event B

Figure 3: Voltage magnitude plots across all buses during Events A and B

Taken together, these signals suggest that the datasets capture both physical fault scenarios and cyber-induced anomalies. Event A likely represents a traditional fault event that prompted both a physical disturbance and a moderate cyber response. In contrast, Event B shows no major physical deviation but triggers an unusually strong and irregular cyber signal pattern, consistent with spoofing or cyber-injected interference. These observations highlight the importance of correlating physical and cyber signals to detect and classify abnormal behavior in power grid systems.

2 ARMA model

3 SARMA model

Bibliography